

TOPIC 2/60

What is Generative AI?

Machines that don't just think.

They create.

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THE BIG SHIFT

For decades, AI was mainly used to **analyze** data. Now, it generates it.

Old AI: The Critic

"This looks like a picture of a dog."

GenAI: The Artist

"Here is a brand new painting of a dog riding a skateboard."

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HOW DOES IT WORK?



Step 1: Ingestion

You feed the AI a massive amount of data. Essentially, a huge chunk of the public internet.

It reads Wikipedia, Reddit, books, code repositories, and articles. It looks at millions of images and their captions.

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THE SECRET SAUCE

Pattern Matching

(Not Magic)

It doesn't actually "understand" anything. It just becomes mathematically incredible at guessing what comes next.

"The sky is ____"

Based on billions of texts it read, it calculates that "blue" is the statistically most likely next word.

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MODALITY 1

Text Gen

Large Language Models (LLMs)

Models trained specifically on text to write essays, code, poems, and emails.

Examples: ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google). They predict text one "token" (part of a word) at a time.

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MODALITY 2

Image Gen

Diffusion Models

Instead of predicting words, these predict pixels.

How Diffusion works:

1. Start with completely random TV static (noise).
2. Gradually remove the noise step-by-step.
3. Guide the process so the final image matches your text prompt (e.g., Midjourney).

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EXPANDING FRONTIERS

AUDIO GENERATION

Cloning voices or generating full musical tracks from text prompts.

> e.g., ElevenLabs, Suno.

VIDEO GENERATION

Creating realistic video physics and scenes purely from imagination.

> e.g., OpenAI Sora, Runway Gen-3.

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THE CATCH

Hallucinations

Because GenAI is built on probability (guessing the next word) and not a strict database of facts, it lies confidently.

DANGER

If the math says a completely fictional fact sounds highly plausible, it will output it as absolute truth. Always verify!

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WHY IT CHANGES EVERYTHING

Zero Marginal Cost

Creating high-quality text or images used to take hours of human labor. Now it takes seconds.

Copilots, Not Replacements

It lowers the barrier to entry. Anyone can code, write, or design with an AI assistant.

Infinite Personalization

Content can be generated on-the-fly, unique to every single user's exact context.

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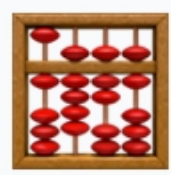


Generative AI

CONCEPT UNLOCKED



Reads the Internet



Learns Patterns



Guesses the Next Token



Generates Novel Content

CONNECT WITH ME



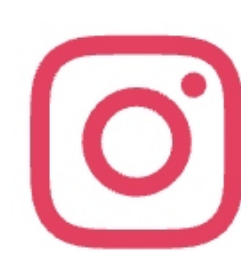
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